

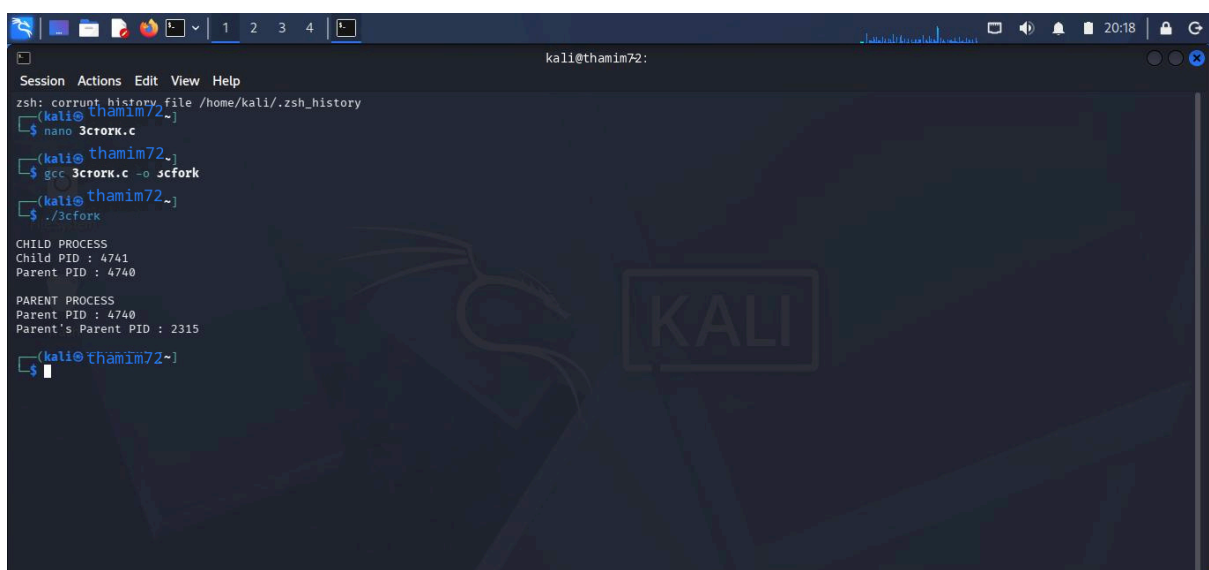
# C PROGRAMMING : fork()



```
GNU nano 9.0 3cfork.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    pid = fork();
    if(pid < 0)
    {
        printf("Fork Failed\n");
        exit(1);
    }
    else if(pid == 0)
    {
        printf("\nCHILD PROCESS");
        printf("\nChild PID : %d", getpid());
        printf("\nParent PID : %d\n", getppid());
        exit(0);
    }
    else
    {
        wait(NULL);
        printf("\nPARENT PROCESS");
        printf("\nParent PID : %d", getpid());
        printf("\nParent's Parent PID : %d\n", getppid());
    }
    return 0;
}
```

## OUTPUT :



```
kali@thamim72:~$ zsh: current history file /home/kali/.zsh_history
kali@thamim72~$ nano 3cfork.c
kali@thamim72~$ gcc 3cfork.c -o 3cfork
kali@thamim72~$ ./3cfork

CHILD PROCESS
Child PID : 4741
Parent PID : 4740

PARENT PROCESS
Parent PID : 4740
Parent's Parent PID : 2315
kali@thamim72~$
```

# SHELL PROGRAMMING :fork()

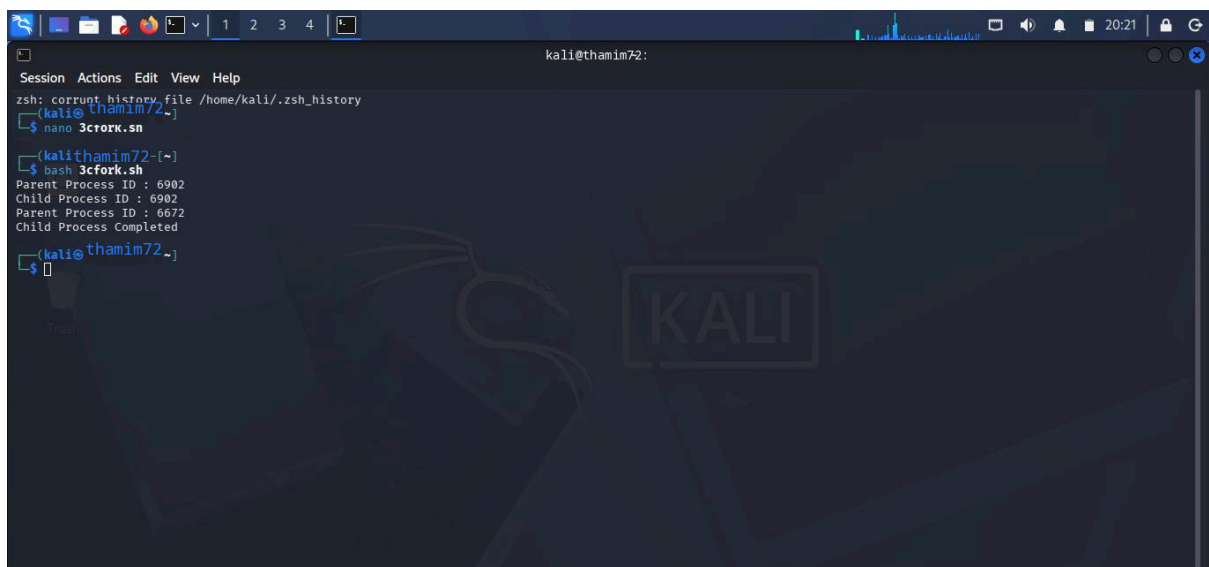


A screenshot of a terminal window on a Kali Linux system. The terminal is running the nano 9.0 text editor, editing a file named 3cfork.sh. The script content is as follows:

```
#!/bin/bash
echo "Parent Process ID : $$"
(
  echo "Child Process ID : $$"
  echo "Parent Process ID : $PPID"
  exit 0
) &
wait
echo "Child Process Completed"
```

The terminal window has a title bar showing "kali@YUVARAJK: ~" and a status bar at the bottom with various keyboard shortcuts like "Help", "Exit", "Write Out", "Read File", etc. The background of the terminal features a Kali Linux logo and the word "KALI".

## OUTPUT :

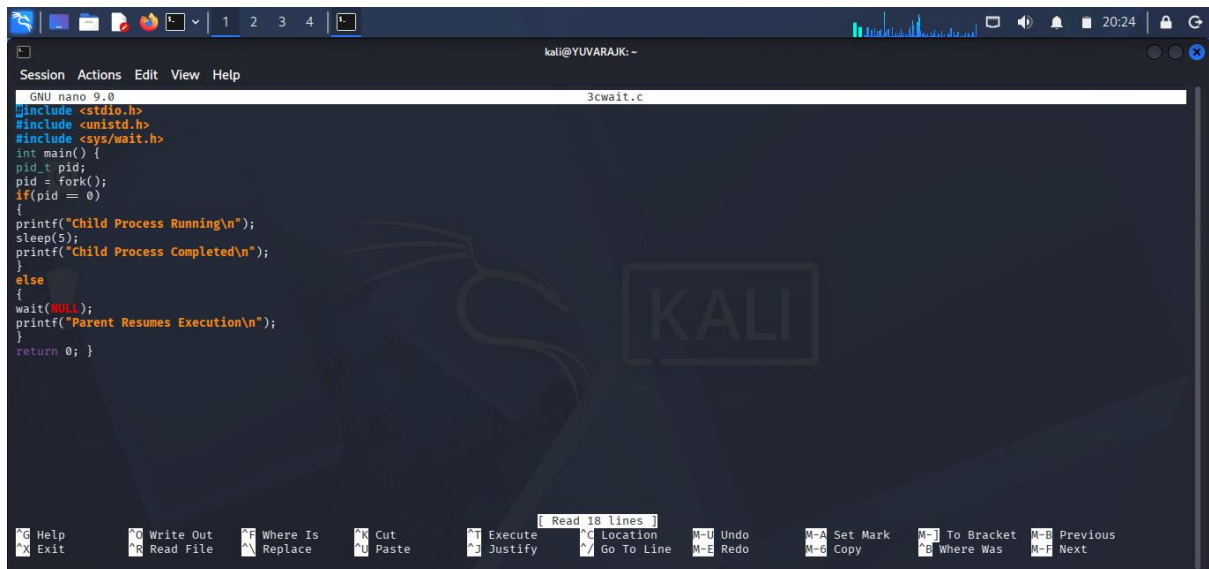


A screenshot of a terminal window on a Kali Linux system. The terminal shows the execution of the 3cfork.sh script. The output is as follows:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano 3cfork.sh
(kali@thamim72~)
$ bash 3cfork.sh
Parent Process ID : 6902
Child Process ID : 6902
Parent Process ID : 6672
Child Process Completed
(kali@thamim72~)
$
```

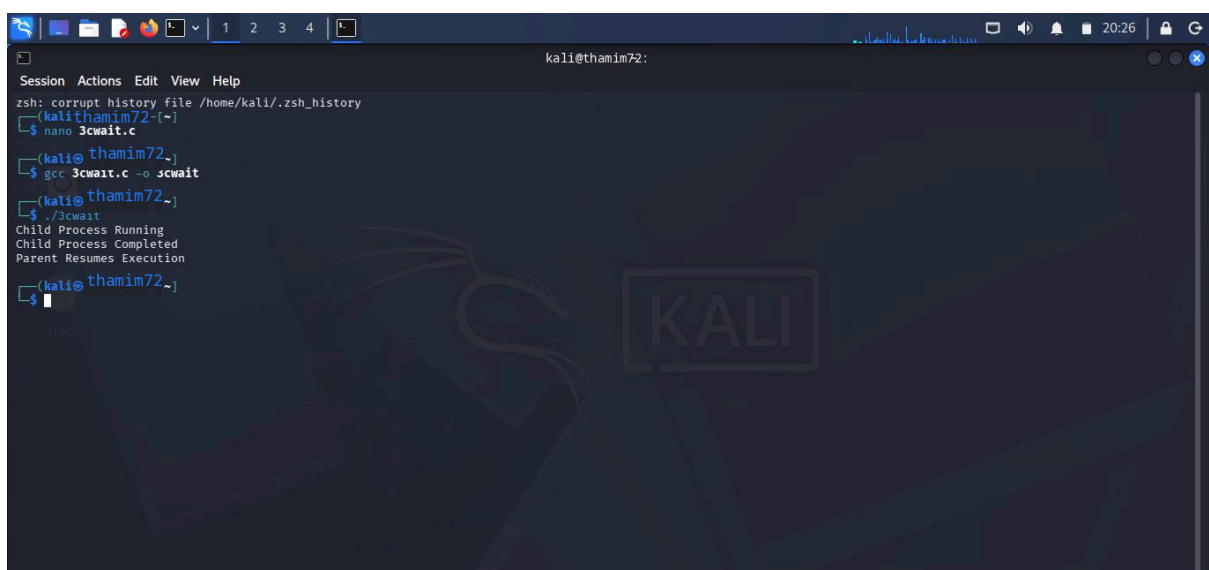
The terminal window has a title bar showing "kali@thamim72:" and a status bar at the bottom with various keyboard shortcuts like "Help", "Exit", "Write Out", "Read File", etc. The background of the terminal features a Kali Linux logo and the word "KALI".

## C PROGRAMMING :wait()



```
GNU nano 9.0 3cwait.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
    pid_t pid;
    pid = fork();
    if(pid == 0)
    {
        printf("Child Process Running\n");
        sleep(5);
        printf("Child Process Completed\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Resumes Execution\n");
    }
    return 0; }
```

## OUTPUT :



```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@thamim72:~$ nano 3cwait.c
kali@thamim72:~$ gcc 3cwait.c -o 3cwait
kali@thamim72:~$ ./3cwait
Child Process Running
Child Process Completed
Parent Resumes Execution
kali@thamim72:~$
```

# SHELL PROGRAMMING :wait()



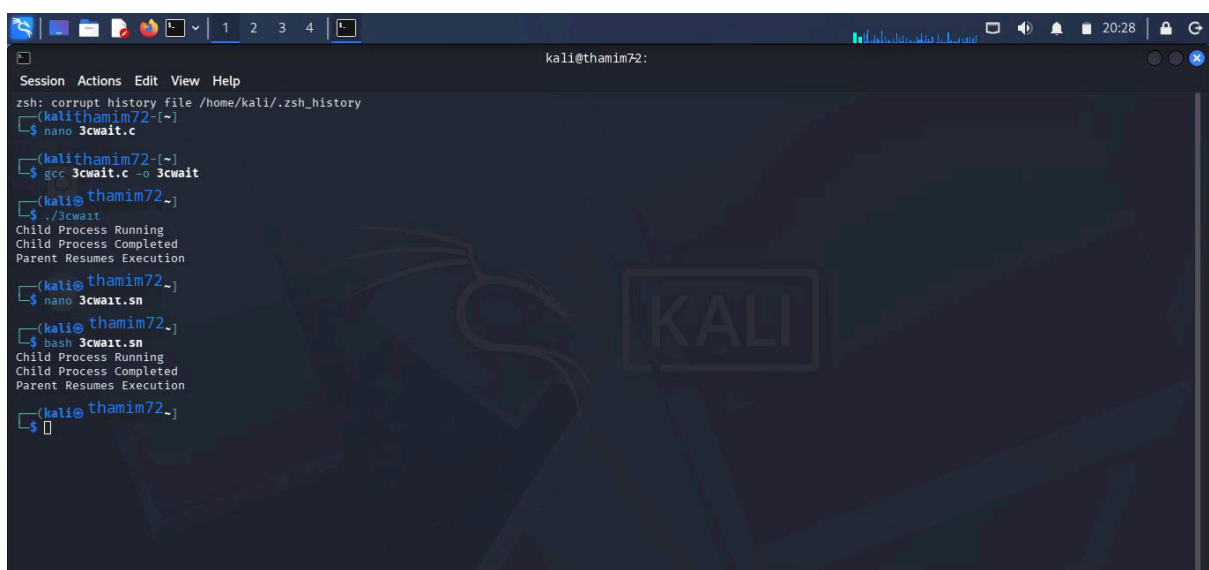
```
GNU nano 9.0 3cwait.sh
#!/bin/bash

(
    echo "Child Process Running"
    sleep 5
    echo "Child Process Completed"
) &

wait

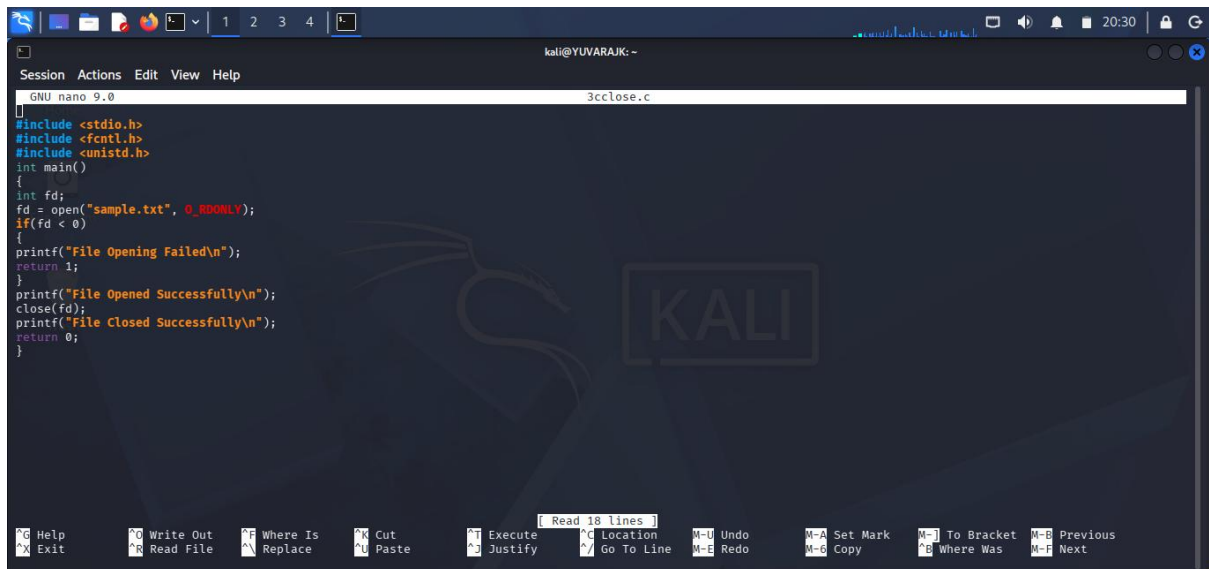
echo "Parent Resumes Execution"
```

## OUTPUT :



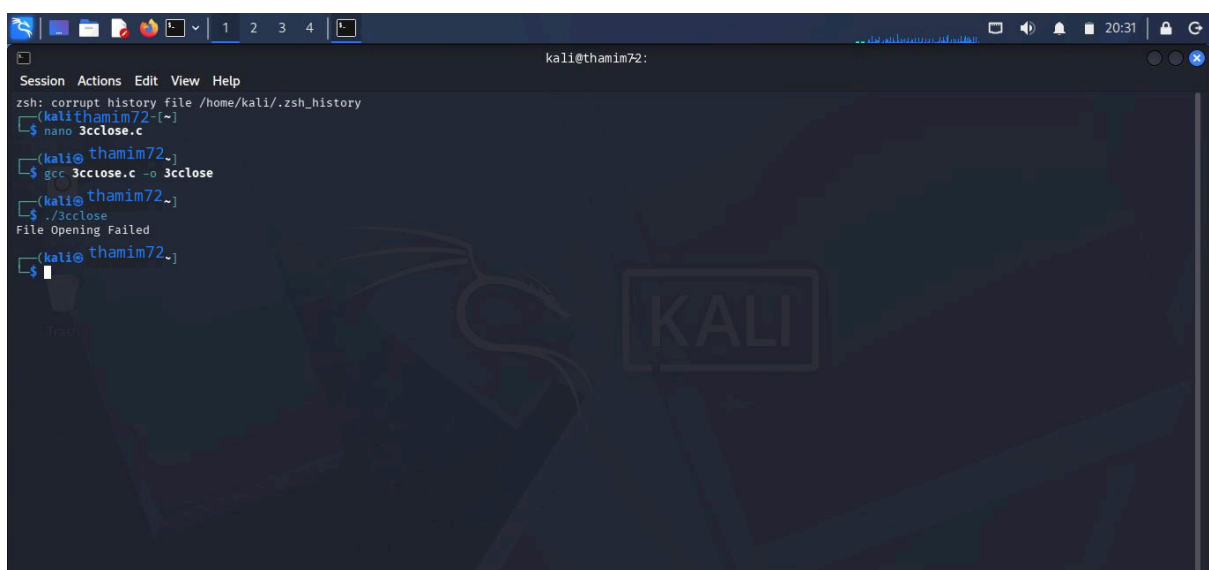
```
kali@thamim72:~
zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72:~)
$ nano 3cwait.c
(kali@thamim72:~)
$ gcc 3cwait.c -o 3cwait
(kali@thamim72:~)
$ ./3cwait
Child Process Running
Child Process Completed
Parent Resumes Execution
(kali@thamim72:~)
$ nano 3cwait.sn
(kali@thamim72:~)
$ bash 3cwait.sn
Child Process Running
Child Process Completed
Parent Resumes Execution
(kali@thamim72:~)
$
```

# C PROGRAMMING :close()



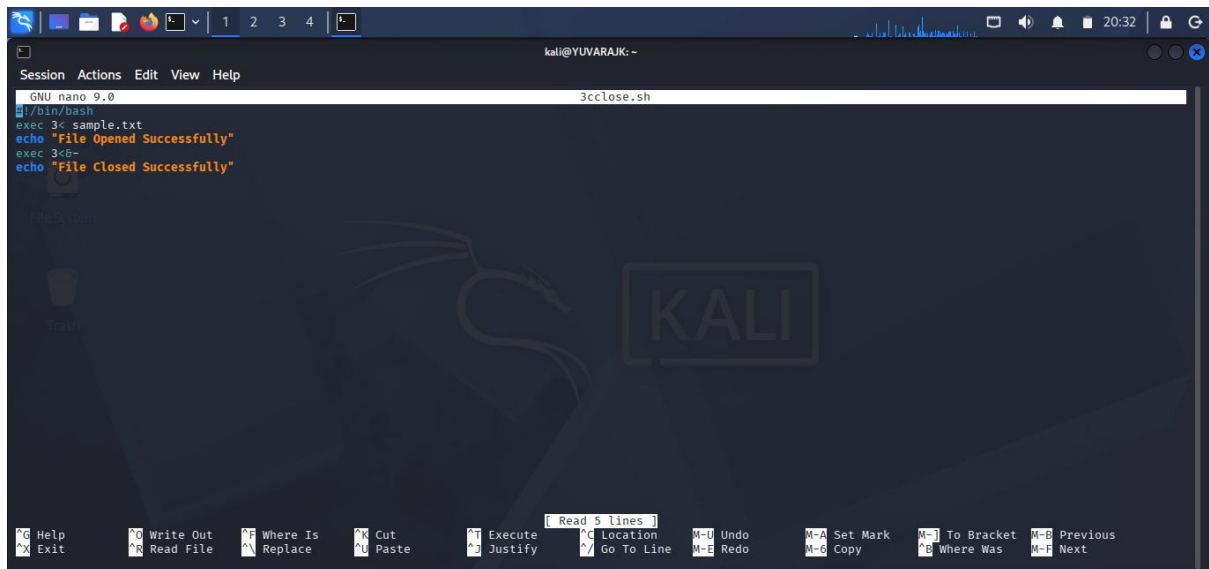
```
GNU nano 9.0 3cclose.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd;
    fd = open("sample.txt", O_RDONLY);
    if(fd < 0)
    {
        printf("File Opening Failed\n");
        return 1;
    }
    printf("File Opened Successfully\n");
    close(fd);
    printf("File Closed Successfully\n");
    return 0;
}
```

## OUTPUT :



```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@thamim72:~$ nano 3cclose.c
kali@thamim72:~$ gcc 3cclose.c -o 3cclose
kali@thamim72:~$ ./3cclose
File Opening Failed
kali@thamim72:~$
```

# SHELL PROGRAMMING :close()

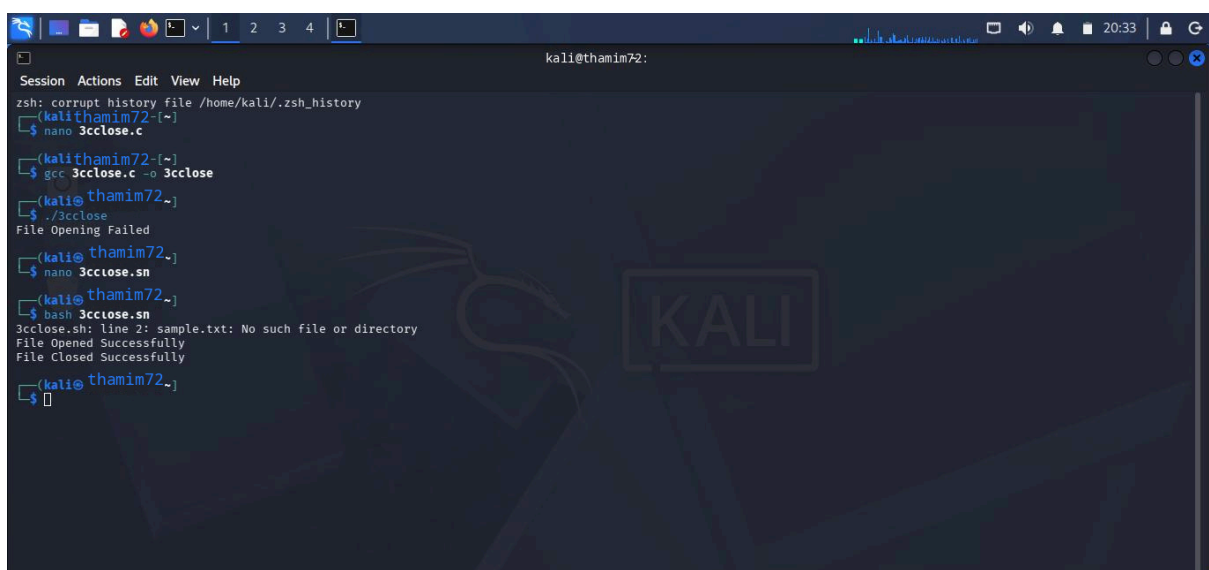


```
GNU nano 9.0 3cclose.sh
#!/bin/bash
exec 3< sample.txt
echo "File Opened Successfully"
exec 3<&-
echo "File Closed Successfully"
```

Read 5 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next

## OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72:~)
$ nano 3cclose.c
(kali@thamim72:~)
$ gcc 3cclose.c -o 3cclose
(kali@thamim72:~)
$ ./3cclose
File Opening Failed
(kali@thamim72:~)
$ nano 3cclose.sh
(kali@thamim72:~)
$ bash 3cclose.sh
3cclose.sh: line 2: sample.txt: No such file or directory
File Opened Successfully
File Closed Successfully
(kali@thamim72:~)
$
```